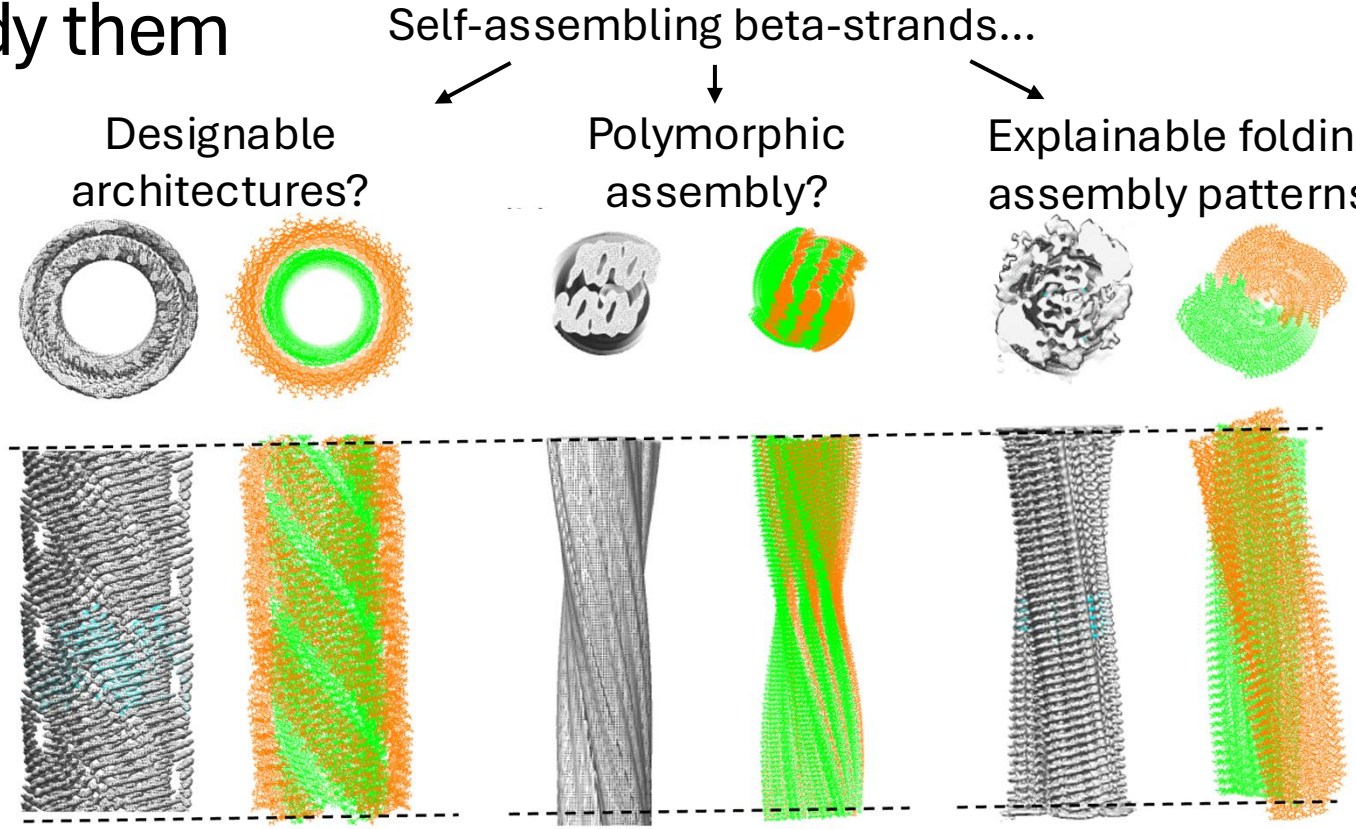
FibrilGen: a Python package to generate cross-beta nanostructures

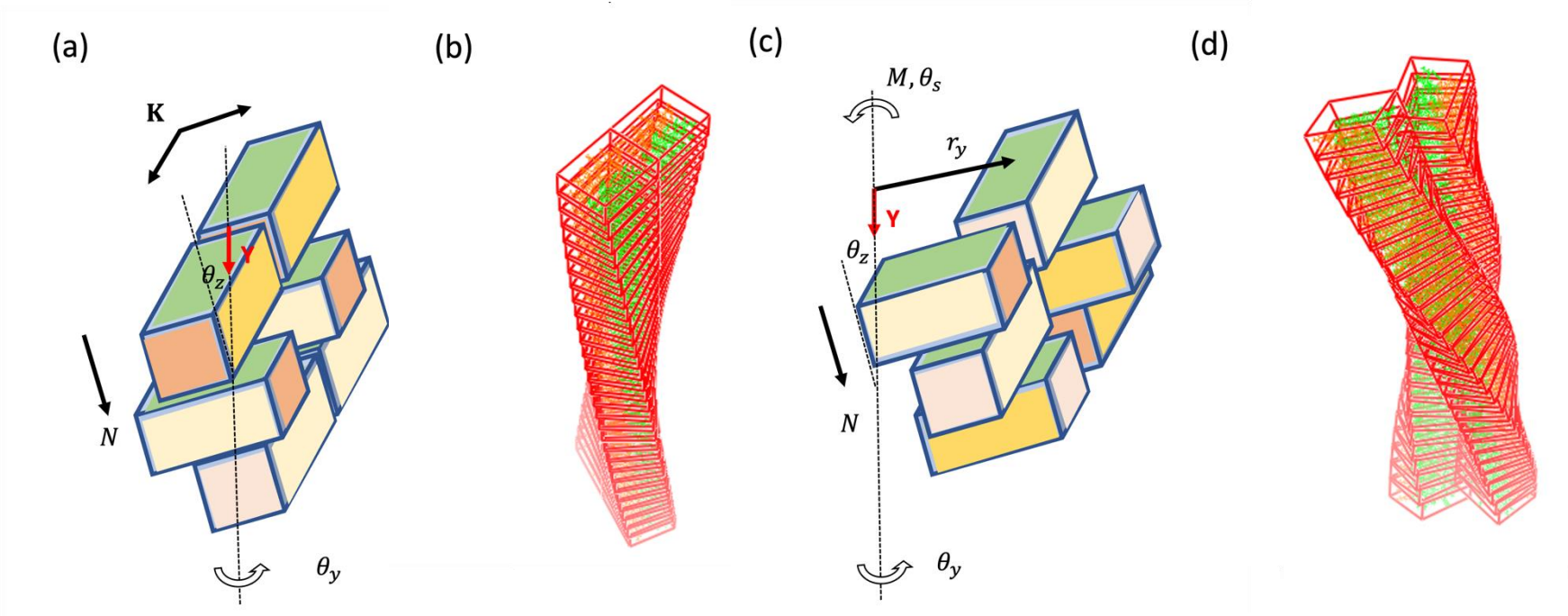
Presenter: Chao-Yu Yang

What are cross-beta nanostructures and why we want to study them

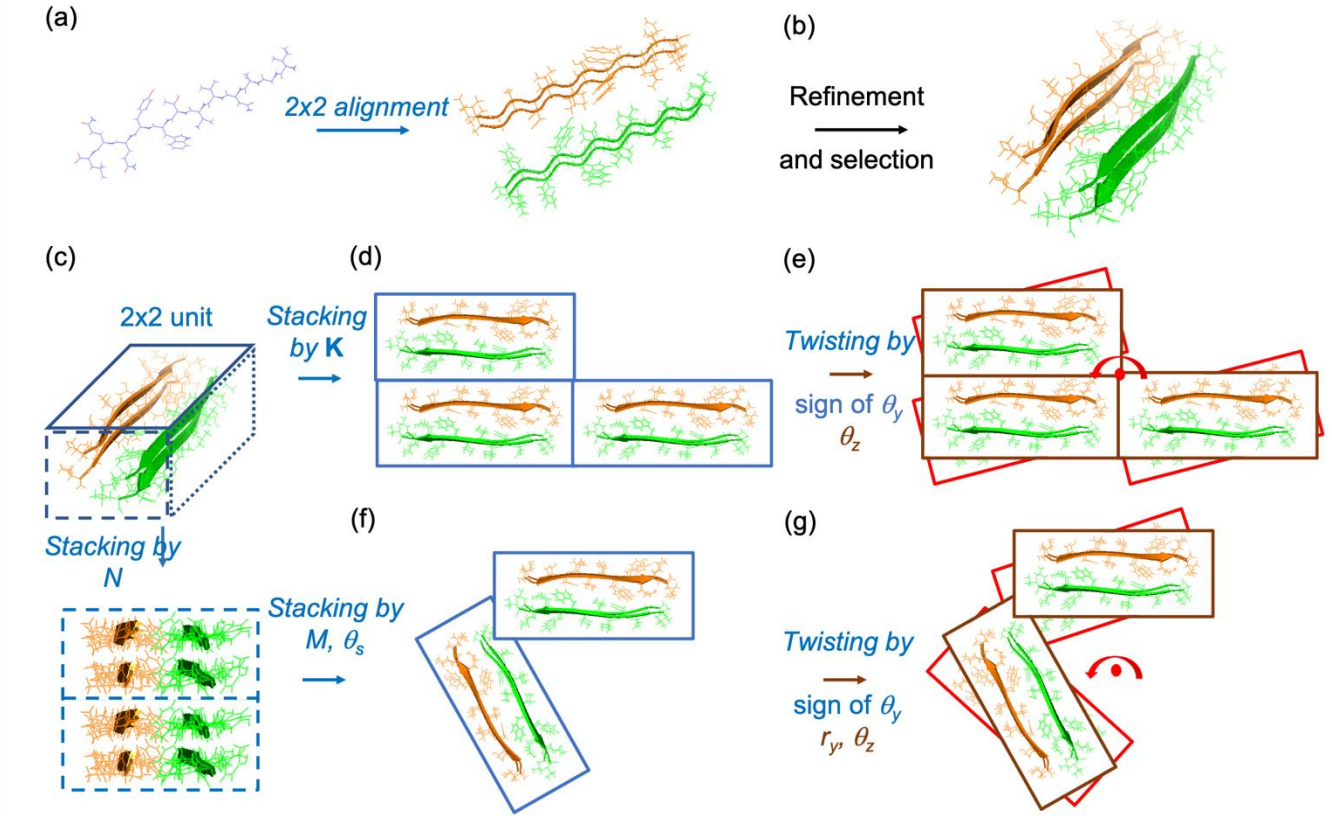


Wang, F. et al. (2021), Schmidt, A. et al. (2016), Gremer, L. et al. (2017), Yang, C.-Y. et al. (2025)

An overview of common cross-beta geometry

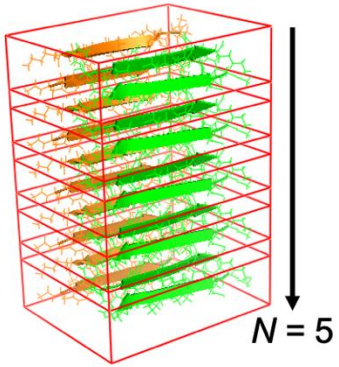


Assemble a peptide molecule into a cross-beta structure



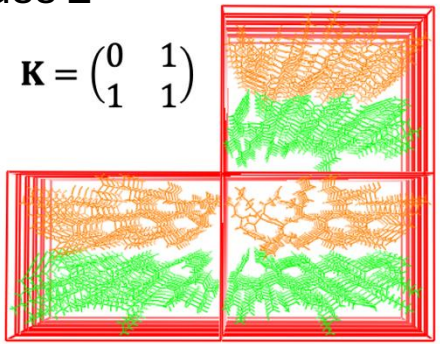
Seven parameters for six classes of cross-beta models

Class 1



Class 2

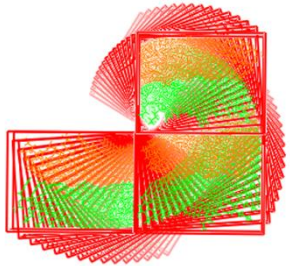
$$\mathbf{K} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$



Class 3



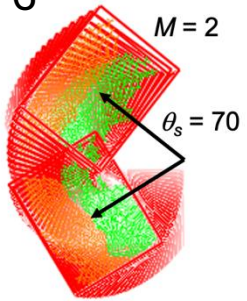
Class 4



Class 5



Class 6



Let us build cross-beta structures!

Peptides, periodic building blocks, nanofibrils and more …

Setup a Colab working environment

- Colab file: https://github.com/ChaoYuYang0/FibrilGen-v1/blob/main/demo_Colab.ipynb

Setup a Python environment

First of all, we setup an environment with essential Python libraries.

```
[ ]
!pip install pymol-open-source==3.2.0a0
!pip install openbabel==3.2.1
!pip install py3Dmol==2.5.5
!git clone https://github.com/ChaoYuYang0/FibrilGen-v1
```

Change the working directory

Once we finish the setup, we change the working directory to "FibrilGen-v1".

```
[ ]
%cd FibrilGen-v1/
```



Build a beta-strand

Basic

Input: a peptide sequence (one-letter code format)

Output: a beta-strand (PDB format)

Advanced

How do side–chain rotations create steric clashes?

Interface

Step 1: build your beta-strand

In this section, we change the input peptide sequence and visualise the output peptide molecule. We change the side-chain rotation angles to see how they can introduce steric clashes (optional).

[ ]

Peptide sequence

Side-chain rotation angles

Input_seq

FEFKFEFK

Angle_sidechain

1_n_ca_cb_cg_-20,3_n_ca_cb_cg_-20,5_n_ca_cb_cg_-20,7_n_ca_cb_cg_-20

Ouput_filename

F8.pdb

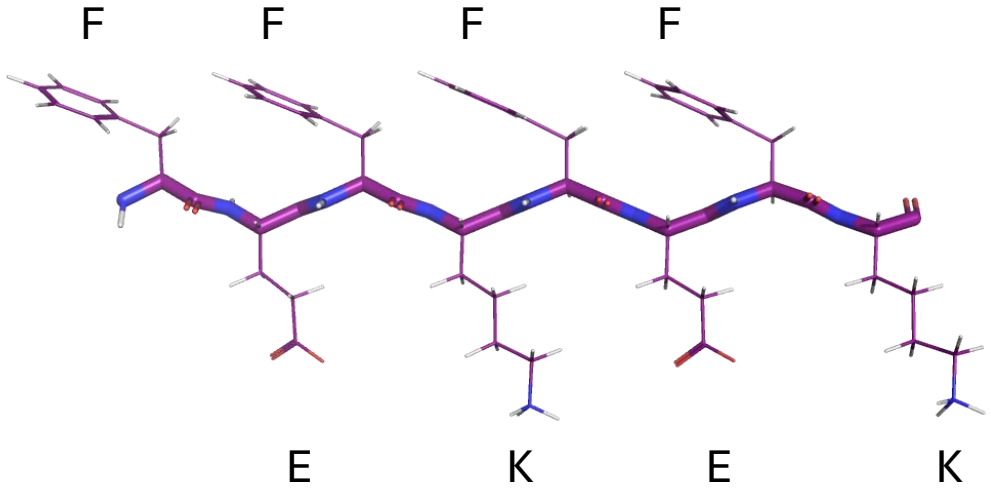
Show_the_structure

Show code

Side-chain rotation angles written as: residx_atom1_atom2_atom3_atom4_angle1, ...

Practice

1. Run the example
2. Try a different peptide sequence
3. Try a different side-chain rotation angle





Build a bilayer structure

Basic

Input: a peptide molecule (PDB format), peptide alignment for a 2 x 2 unit

Output: an energy minimised beta-sheet (PDB format)

Advanced

How peptide molecule and peptide alignment destabilise the beta-sheet assembly?

Interface

Step 2: Build your bilayer structure

In this section, we change the input peptide alignment and visualise the output 2 x 2 beta-sheet building block. We multiply the 2 x 2 building block 5 times, then optimise the assembled structure. We check whether inter-peptide backbone hydrogen bonds and side-chain packing are plausible in the assembled structure.

[ ]

Input_filename: F8.pdb

Num_residues_per_peptide: 8

Reference_segment: 2-8

Alignment_pep_orientations: aaa

Output_filename: F8_bilayer.pdb

Start_res_at_inner_face

Alignment_face_to_face

Energy_minimisation

Show_the_structure

Show code

Peptide molecule

Linear segment to align

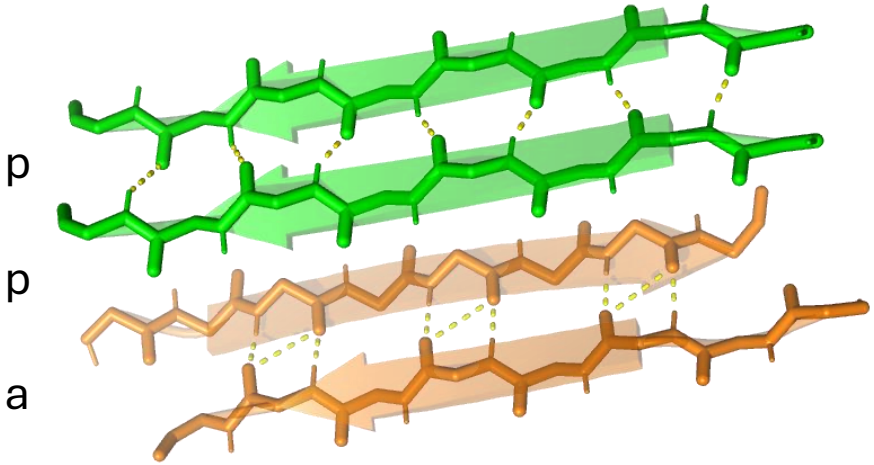
Backbone alignment

Face side

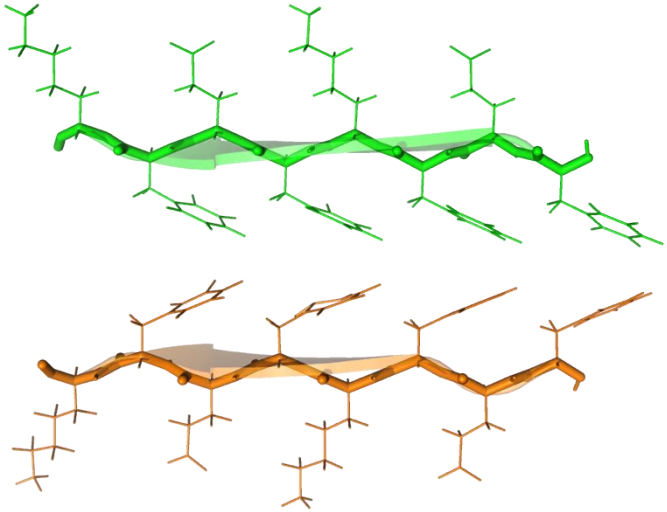
Inter-sheet face alignment

Practice

1. Run the example
2. Try a different peptide molecule
3. Try a different peptide alignment



Aligned face-to-face





Build a fibril structure

Basic

Input: reference bilayer (PDB format), input parameters for a fibril model

Output: a cross-beta structure (PDB format)

Advanced

How do 2 x 2 units affect the feasible assembly of beta-sheet structure?

Interface (class 1 **sheet** structures)

Step 3: Build your fibril structure

In this section, we use a 2 x 2 building block to assemble different classes of cross-beta structures.

(a) We use our optimised bilayer to assemble a **sheet** structure. We change the number of stacking units (parameter N) along the beta-sheet axis and visualise the structure. The output structure is built from stacked unit blocks.

Reference bilayer

FibrilGen parameters

[ ]

Input_filename: F8_bilayer.pdb

Input_num_peptides_per_sheet: 10

Num_residues_per_peptide: 8

Reference_segment: 2-8

N: 10

Output_filename: F8_sheet.pdb

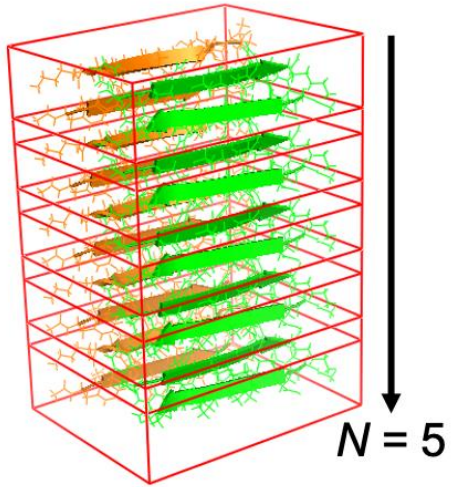
Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.

Show code

Practice (class 1 **sheet** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.



Interface (class 2 **stacked sheet** structures)

(b) We use our optimised bilayer to assemble a **stacked sheet** structure. We vary the number of stacking units in the sheet cross-section (parameter **K**; do not specify element 1 surrounded by element 0) and the number of stacking units (parameter N) along the beta-sheet axis. The output structure is built from stacked unit blocks.

[ ]

Reference bilayer

FibrilGen parameters

Input_filename: F8_bilayer.pdb

Input_num_peptides_per_sheet: 10

Num_residues_per_peptide: 8

Reference_segment: 2-8

N: 10

K: 0_1_0,1_1_1

Output_filename: F8_stacked_sheet.pdb

Show_the_structure

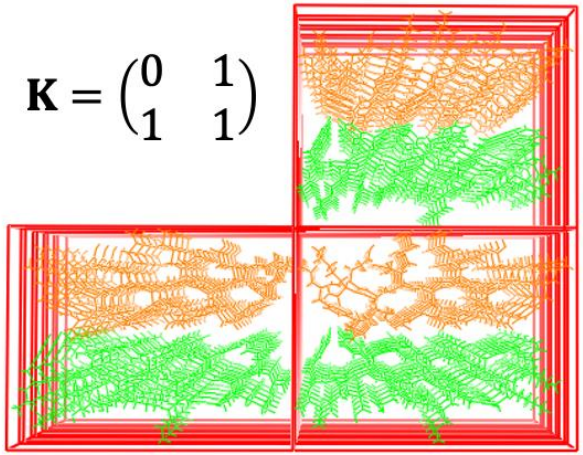
- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter **K** is a binary matrix ("_" to seperate columns, "," to sperate rows) assigning whether a 2 x 2 unit is presented.

Show code

Practice (class 2 **stacked sheet** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different **K**. Increase the number of presented units for a thicker fibril.

$$\mathbf{K} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$



Interface (class 3 **rod** structures)

(c) We use our optimised bilayer to assemble a **rod** structure. We change the twist chirality (the sign of parameter θ_y), the initial tilt angle (parameter θ_z), and the number of stacking units (parameter N) along the fibril axis. The output structure is built from stacked and twisted unit blocks.

We provide tips for parameter assignments:

- To relax the structure, try a smaller tilt angle θ_z.
- The tilt angle is automatically refined to avoid steric clashes. The initial value affects the helical twist it converges to (or fails to converge to).

[ ]

Reference bilayer

FibrilGen parameters

Input_filename	Input_num_peptides_per_sheet	Num_residues_per_peptide
F8_bilayer.pdb	10	8
Reference_segment	**N**	**theta_z**
2-8	50	5.0
Sign_theta_y	**Output_filename**	☑ Show_the_structure
1	F8_rod.pdb	

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter θ_z is the initial tilt angle.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Show code

Practice (class 3 **rod** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
4. Try a different θ_z. Increase the number for a shorter fibril.



Interface (class 4 **stacked rod** structures)

(d) We use our optimised bilayer to assemble a **stacked rod** structure. We change the number of stacking units on the fibril cross-section (parameter **K**; do not specify element 1 surrounded by element 0), the twist chirality (the sign of parameter θ_y), the initial tilt angle (parameter θ_z), and the number of stacking units (parameter N) along the fibril axis. The output structure is built from stacked and twisted unit blocks.

We provide tips for parameter assignments:

- To relax the structure, try a smaller tilt angle θ_z.
- The tilt angle is automatically refined to avoid steric clashes. The initial value affects the helical twist it converges to (or fails to converge to).

Reference bilayer

FibrilGen parameters

[ ]

Input_filename	Input_num_peptides_per_sheet	Num_residues_per_peptide
F8_bilayer.pdb	10	8
Reference_segment	**N**	**K**
2-8	10	0_1,1_1
theta_z	**Sign_theta_y**	**Output_filename**
10.0	1	F8_stacked_rod.pdb

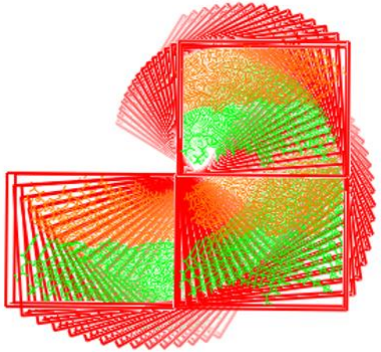
Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter **K** is a binary matrix ("_" to seperate columns, "," to sperate rows) assigning whether a 2 x 2 unit is presented.
- Parameter θ_z is the initial tilt angle.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Show code

Practice (class 4 **stacked rod** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different **K**. Increase the number of presented units for a thicker fibril.
4. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
5. Try a different θ_z. Increase the number for a shorter fibril.



Interface (class 5 **ribbon** structures)

(e) We use our optimised bilayer to assemble a **ribbon** structure. We change the twist chirality (the sign of parameter θ_y), the initial tilt angle (parameter θ_z), the initial radius of rotation (parameter r_y), and the number of stacking units (parameter N) along the fibril axis. The output structure is built from stacked and twisted unit blocks.

We provide tips for parameter assignments:

- To relax the structure, try a smaller tilt angle θ_z.
- The tilt angle and radius are automatically refined to avoid steric clashes. The initial values affect the helical twist they converge to (or fail to converge to).

Reference bilayer

FibrilGen parameters

[ ]

Input_filename	Input_num_peptides_per_sheet	Num_residues_per_peptide
F8_bilayer.pdb	10	8
Reference_segment	**N**	**theta_z**
2-8	50	30.0
r_y	**Sign_theta_y**	**Output_filename**
60.0	-1	F8_ribbon.pdb

Show_the_structure

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter θ_z is the initial tilt angle.
- Parameter r_y is the initial radius of rotation.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Show code

Practice (class 5 **ribbon** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
4. Try a different θ_z. Increase the number for a shorter fibril.
5. Try a different r_y. Increase the number for a wider fibril.



Interface (class 6 **stacked ribbon** structures)

(f) We use our optimised bilayer to assemble a **stacked ribbon** structure. We change the number of stacking units (parameter M) and the angle for stacking (parameter θ_s) on the fibril cross-section; the twist chirality (the sign of parameter θ_y); the initial tilt angle (parameter θ_z); the initial radius of rotation (parameter r_y); and the number of stacking units (parameter N) along the fibril axis. The output structure is built from stacked and twisted unit blocks.

We provide tips for parameter assignments:

- Assign θ_s and M such that $\theta_s \cdot M < 360$ degrees.
- To relax the structure, try a smaller tilt angle θ_z.
- The tilt angle and radius are automatically refined to avoid steric clashes and adjust the distance between neighbouring beta-sheets. The initial values affect the helical twist they converge to (or fail to converge to).

Reference bilayer

FibrilGen parameters

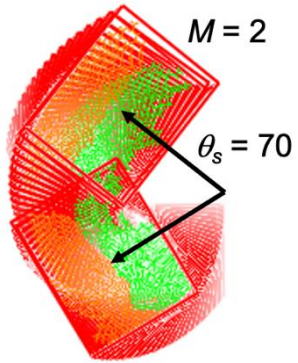
[ ]

Input_filename	Input_num_peptides_per_sheet	Num_residues_per_peptide
F8_bilayer.pdb	10	8
Reference_segment	**N**	**theta_s**
2-8	20	70.0
M	**theta_z**	**r_y**
3	10.0	30.0
Sign_theta_y	**Output_filename**	☑ Show_the_structure
1	F8_stacked_ribbon.pdb	

- Parameter N is an integer repeating stacking along the vertical axis.
- Parameter θ_s is angular displacement of units on the fibril cross-section.
- Parameter M is the number of stacking on the fibril cross-section.
- Parameter θ_z is the initial tilt angle.
- Parameter r_y is the initial radius of rotation.
- Parameter "sign of θ_y"" assigned to 1 for left-handed twist, and -1 for right handed twist.

Practice (class 6 **stacked ribbon** structures)

1. Run the example
2. Try a different *N*. Increase the number for a longer fibril.
3. Try a different sign of θ_y. Assign 1 for a left-handed twist and -1 for a right-handed twist.
4. Try a different θ_z. Increase the number for a shorter fibril.
5. Try a different r_y. Increase the number for a wider fibril.
6. Try a different *M*. Increase the number of presented units for a more enclosed fibril.
7. Try a different θ_s. Increase the number for a more twisted fibril.



Enjoy building cross-beta fibrils?

We welcome feedback and suggestions!